

Bachelor Degree Project



Stress and Coping in Sweden and Sri Lanka

A cross-cultural study with a cognitive
neuroscientific perspective

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Abstract

The stress response that is triggered in an organism when facing a stressor is crucial to maintain stability and health. However, exposure to a severe or a chronic stressor can be maladaptive and cause several impairments in the body, such as cardiovascular diseases, atrophy of the brain, and psychopathologies mainly characterized by anxiety and depression. Resilience or vulnerability to stress is mediated through different biopsychosocial factors, one of which is the use of coping strategies. Different types of coping strategies have been linked to either adaptive or maladaptive outcomes, and are an important factor to consider regarding stress resilience. Cultural differences in symptoms of stress, anxiety, depression, and coping strategies were assessed through self-report measurements in the form of two questionnaires. 75 Swedes and 67 Sri Lankans between the age of 18-50 took part in the study. The most significant findings of this study suggest that 1) Sri Lankans experience more symptoms of stress, anxiety, and depression compared to Swedes, 2) dysfunctional coping is correlated with higher levels of stress, anxiety, and depression in both Sweden and Sri Lanka, 3) higher levels of stress predicts higher levels of anxiety and depression in both Sweden and Sri Lanka, and 4) both countries tend to favor problem-focused coping over emotion-focused and dysfunctional coping. A discussion regarding the current findings, including limitations of the study is provided, as well as directions for future research.

Keywords: stress, major depressive disorder, posttraumatic stress disorder, suicide, resilience, coping, cultural differences, Sri Lanka

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Introduction

"It is not stress that kills us, it is our reaction to it"

– Hans Selye

The foundation of our modern concept of stress was laid in the mid 19th century, by a physiologist named Claude Bernard. Bernard argued that all living things have an internal environment that is regulated to withstand and adjust to changes in the external environment, and that the maintenance of life is dependent on this mechanism (Lovallo, 2016). A *stressor* is a change in the external environment that is perceived as a threat, and a *stress response* is the compensative reaction to stabilize the body and maintain health.

Stressors can be categorized as either physical or psychological. Physical stressors, such as heat, injuries, infections, and so on, threaten our physical well being. But what we often refer to when talking about stress in our everyday lives are psychological stressors, which threatens our well-being indirectly (Lovallo, 2016). Such stressors could be solving a mathematical problem, missing the bus, or getting fired from your job. These kind of stressors are often not life-threatening. However, if they remain chronic, they can have serious effects on the body. These negative consequences are mainly caused by the body's own stress response mechanisms (McEwen, 2000).

Hans Selye, another major figure in the history of the concept of stress, discovered that the stress response causes long-term changes in the body (Lovallo, 2016). A normally functioning stress response leads in first hand to adaptive changes in the body, which are crucial to keep an organism alive (McEwen, 2000). However, the adaptive effects can become maladaptive when the mediators of the stress response malfunction through either a prolonged

response that does not shut off even when the stressor is over or a response that is not turned on to the extent that the situation demands (McEwen, 2003). If the stress response remains maladaptive over time, it can become pathological and cause serious impairments in the body (McEwen, 2003). Such pathologies are for example Posttraumatic Stress Disorder (PTSD) and Major Depressive Disorder (MDD), which at their most severe, have been linked to suicide attempts (Beautrais, 2000).

Suicide is a worldwide health problem, accounting for approximately 800 000 deaths every year, and was the second most common cause of death between 15-29-year-olds in 2015 (WHO, 2015). According to the latest data, Sri Lanka has the highest suicide rates in the world with 35.3 per 100 000 population (WHO, 2015). To put this rate in perspective, the global average of suicide mortality rate is 10.7 per 100 000, and furthermore, Sweden had in 2015 a suicide rate of 15.4 per 100 000 compared to Sri Lanka (WHO, 2015).

The importance of reducing suicides speaks for itself. It can be argued that every human being should be able to have a life worth living. A key first step to approach this health problem could be through identifying crucial risk-factors for committing suicide. By looking at previous research concerning main risk factors for suicide in both western and eastern countries, it can be noted that one common denominator is stress. In western countries, mood disorders like MDD and PTSD have been strongly linked to suicide attempts (Kendler, Karkowski, & Prescott, 1999; Mazure, 1998; Slewa-Younan et al., 2017), and many studies shows that severe or chronic stress is often the leading cause of the development of these conditions (Beautrais, 2000; Bridge, Goldstein, & Brent, 2006; Cavanagh, Carson, Sharpe, & Lawrie, 2003; Conwell, Duberstein, & Caine, 2002; Gould, Greenberg, Velting, & Shaffer, 2003). In eastern countries, including Sri Lanka, direct exposure to stress in the form of severe life events, such as death of a close relative

of friend, or marital disharmony, seems to be, together with mental illness, the most common risk factors for suicide attempts (Beautrais, 2006; Pearson et al., 2014; Samaraweera, Sumathipala, Siribaddana, Sivayogan, & Bhugra, 2008).

Thus, it could be suggested that a possible way of trying to reduce suicide is through understanding the mechanisms of stress that tend to trigger psychopathology, and how to build resilience in order to minimize these negative outcomes. Previous research has linked several different biopsychosocial factors to stress resilience, and the use of coping strategies have been suggested to play an important role. As research has shown that a worrying feature of suicide risk in Sri Lanka involves mental illness and a difficulty to cope with severe life stressors, it has been proposed that good attempts to reduce suicide in the country could involve improving problem-solving and coping skills, and treatment for mental illness (Beautrais, 2006). To the best of the author's knowledge, a study concerning the relationship between the use of different coping strategies and symptoms of stress and stress-related psychopathology has yet not been conducted in Sri Lanka. Identification of maladaptive coping strategies and high levels of stress, anxiety and depression symptoms could provide a greater understanding of the underlying factors to the high suicide rates in Sri Lanka, and give an indication of possible interventions.

The current thesis is concerned with several aims. First, a theoretical overview concerning the mechanisms of stress in the body and stress-induced pathology will be presented, as well as characteristics of resilience to stress. These issues will primarily be dealt with a cognitive neuroscientific approach, with influences of a psychosocial perspective. As a part of stress resilience, the effects of different types of coping strategies will be discussed. Second, an empirical cross-cultural comparison will be provided. Differences in symptoms of stress, anxiety, depression, and use of different coping strategies between a Swedish and Sri Lankan sample will

be conducted and analyzed, as well as the relationship between the use of different types of coping strategies and symptoms of stress, depression, and anxiety.

Definitions of the concept of stress will firstly be outlined, followed by a neuroscientific presentation of the stress response. Protective and damaging effects of mediators of the stress response will then be explained, including stress-induced MDD and PTSD. Moreover, factors of resilience to stress will be presented together with coping strategies. After this theoretical background, the method of the current study will be given, followed by the results and a discussion of the findings. The thesis will end with a concluding remark.

Stress

Definitions of Stress

Stress has been defined as a physical or mental tension being a threat to an organism (Lovallo, 2016). This tension requires a bodily response to compensate for the threat. The greater the stressor is, the greater the stress response has to be to be able to compensate and bring back the body to a stable state, e.g. *homeostasis*. The state of stress is not removed until the compensatory reactions have eliminated the threat. If the stressor is too great or the resources of the organism are inadequate to maintain homeostasis, the organism can be subject to damage or even death (Lovallo, 2016).

One of Hans Selye's largest contributions to the research on stress was noticing that regardless of the stressor, the biological stress response is rather similar (Lovallo, 2016). His research on rats showed that the stress response consists mainly of three components; decreased size in immune system organs, ulcers in the gastrointestinal tract, and an increase in the size of the adrenal glands. Selye argued that these three components form a universal pattern of the

stress response, and he called it *the general adaptation syndrome* (Lovallo, 2016). Except for these three components included in the general adaptation syndrome, he also identified the process of the stress response through which the organism aims for survival. This process consists of three phases. The first one is the *alarm reaction*. The stressor is sensed by the organism and a stress response starts to arise. The second phase is the *stage of resistance*, where the organism produces the necessary chemical or behavioral changes in the body to fight the stressor. This phase continues until the stressor is eliminated, or until the organism has no more resources to resist the stressor. If the organism fails to compensate for the stressor, it is subject to the third and final phase: the *stage of exhaustion*, which at its most severe results in the organism's death (Lovallo, 2016).

As has been mentioned, Selye also discovered that stress causes long-term changes in the body, which can be both adaptive and maladaptive. The negative aspect he found was that when the organism is in the stage of resistance, adding a second stressor, even if it is minor, will cause impairment in the stress response. Also, he found that after a prolonged stressor, the organism's resources remained low for a period of time. The positive aspect he found was that repeated exposure to a mild stressor can actually increase the ability to deal with a more severe and prolonged exposure to the same stressor (Lovallo, 2016).

There is not an either/or relationship between adaptive and maladaptive stress responses. The responses are rather adaptive at some levels, and maladaptive at others (Christopher, 2004). Several factors influence this dynamic relationship, which includes both situational and individual differences. Individual differences are mediated by biopsychosocial factors, such as genetic disposition, biochemical mechanisms, psychological processes, and sociocultural environment (Christopher, 2004). These influence how a specific individual perceives, interprets

and reacts to a stressor. Hence, one individual can have a different reaction to the same stressor at different times, and also, a situation perceived as a stressor for one individual may not be perceived as one by another (Christopher, 2004). Situational differences can make a stress response that is adaptive in one situation be maladaptive in another. The consequences of chronic and severe stress will be further discussed, but for now, what can be concluded is that exposure to stress can lead to long-term consequences, and these can be both adaptive and maladaptive (Lovullo, 2016). The pathway from the perception of a threat to physical stress responses will be presented next.

Biology of The Stress Response

The brain is the most important organ in terms of stress. It is the brain that detects the stressor and interprets situations as threats or not. It also determines the adequate response to the given stressor, both physiological and behavioral (McEwen, 2000). There are many changes in the nervous, endocrine, cardiovascular, and immune systems after detecting a stressor. These changes make up the stress response. The two-way communication between the brain and targeted systems in the body during the stress response is mediated via the nervous and endocrine systems (Schneiderman, Ironson, & Siegel, 2008).

The nervous and endocrine systems. The nervous system consists of the brain and spinal cord, which together form the *central nervous system*, and the nerves that exit the spinal cord and extend to several parts of the body, which makes the *peripheral nervous system*. Information that travels within the nervous system is mediated by electrical impulses and is very fast. The central nervous system receives sensory information from the peripheral nervous system and sends motor commands to it (Gazzaniga, Ivry, & Mangun, 2014). The motor

commands received by the peripheral nervous system are a part of one of two subsystems. The *autonomic motor system*, that controls the involuntary muscles, such as the heart and various glands, and the *somatic motor system*, which controls the voluntary muscles. The autonomic motor system regulates the internal organs unconsciously (Gazzaniga et al., 2014). This system can be further divided into two branches, the *sympathetic* and *parasympathetic* branch. These branches either excite the body and prepare it for action, or relax the body. The sympathetic branch is responsible for the excitatory activation of the organs and is hence controlling the physical and unconscious stress response. The parasympathetic branch, on the other hand, is in charge of the resting state. These systems are in constant interplay. To which extent each branch is responsible for the internal state at any given moment is regulated by the hypothalamus, a small neuroendocrine structure situated superior to the brainstem (Gazzaniga et al., 2014).

The endocrine system consists of various glands throughout the body, e.g. organs that produce and release substances in the body. In contrast to the nervous system, the information sent within the endocrine system travels through the secretion of hormones in the bloodstream. The hormones, in turn, attach to the targeted organs by binding to receptors, to regulate their function. This makes the endocrine system slower than the nervous system. The endocrine and nervous systems operate together to maintain stability in the body and are the main mediators of the stress response (Lovallo, 2016).

Neurobiology of the Psychological Stress Response. It was previously stated that there are two types of stressors; physical and psychological. While these stressors elicit the same stress response in the body, they differ in how the response is triggered. Physical stressors are regulated by the hypothalamus and autonomic nervous system and are mainly unconscious. How

psychological events cause physiological stress reactions is more complex and involves several higher brain systems that influence the regulatory functions of the hypothalamus (Lovallo, 2016).

First, it is important to understand the appraisal of psychological stressors, which in turn impact the physiological response. Our thoughts determine how we respond to events that happen around us, and we constantly evaluate the situations we encounter. This is called the *appraisal process* and was first proposed by Lazarus and Folkman in 1984 (Biggs, Brough, & Drummond, 2017). In this process, we put tags on everything we experience as either familiar or unfamiliar, and threatening or non-threatening. In the same way, we also generate behavioral strategies of how to deal with these events. *Primary appraisal* is the evaluation process, where we grade the threat value of each event. Events that are familiar and non-threatening are ignored, and events that are perceived as challenges or threats undergo the *secondary appraisal* - the evaluation of an appropriate behavioral response to the events (Lovallo, 2016). We look at our resources, options, and effectiveness of possible coping strategies before a response is carried out. These advanced activities cannot be carried out by the hypothalamus, but need brain areas with higher cognitive functions that can perceive and classify external events, form emotions, and shape behavioral responses to them (Biggs et al., 2017).

Before the initiation of autonomic and endocrine responses to a psychological stressor, we perceive sensory information from the environment and interpret it. Based on this process, emotions are generated that lay the ground for our stress response (Lovallo, 2016). Sensory information enters the brain via our sense organs. This information first enters the *thalamus*, that directs the information to the primary cortical projection area for each specific sense. Thereafter the information is processed in different *association areas* of the cortex (Lovallo, 2016). Next,

the *limbic lobe* becomes a part of the pathway to the hypothalamus and a possible stress response.

The limbic lobe consists of several structures of the brain that are associated with the classification and motivational evaluation of incoming stimuli. Some of which are the *cingulate gyrus* and *parahippocampal gyrus*, that includes the *hippocampus* and *amygdala* (Lovallo, 2016). The cingulate gyrus receives information from nearby areas of sensory and motor cortex. After the sensory information has passed through association areas, information is passed on to the parahippocampal gyrus, where the information becomes available for both the hippocampus and amygdala. These subcortical structures have complementary functions in evaluating events. The hippocampus is associated with the formation of declarative memories, making it crucial for remembering daily events (Lovallo, 2016). The amygdala, in turn, is involved with both memory and emotion. It is essential for the formation of classical conditioning, where emotions are paired with external events. In this way, we can say that the amygdala is in charge of emotional memories. Hence, the amygdala is fundamental for us to experience and remember situations of emotional significance. Together, the hippocampus and amygdala store familiar experiences along with their emotional meaning. The hippocampus helps us to remember what occurred, and the amygdala helps us to remember what it felt like. Both pieces are of considerable importance in making informed decisions about behavioral responses. Without these functions, we would lose our ability to prevent negative outcomes of our behavior (Lovallo, 2016).

After the hippocampus and amygdala have labeled the event as either familiar or not, and threatening or not, the information is passed on to the *anterior cingulate cortex* (ACC). The ACC collects information about ongoing sensory processes and motor behavior from the cingulate gyrus, and motivational states from the hippocampus and amygdala, and communicates this to

the *prefrontal cortex* (PFC). This coming together of information in the PFC from the limbic system is called *frontal-limbic* connections (Lovallo, 2016). This evaluation of events, concerning meaning, significance, and affective quality, forms the essence of the appraisal process which in turn results in emotional responses and a possible physiological stress response (Lovallo, 2016).

Acute stressors that trigger the fight-or-flight response are mediated through the sympathetic response of the adrenal medulla. The activation of the acute stress response travels through the autonomic nervous system, which makes it very fast. The signal is triggered in the hypothalamus and travels down the brainstem and the spinal cord until it reaches endocrine glands called the *adrenal glands*, which are located on top of the kidneys. When reaching the adrenal glands *epinephrine* and *norepinephrine* are secreted by the adrenal medulla, the center of the adrenal glands, into the bloodstream (Lovallo, 2016). Epinephrine and norepinephrine are substances that act like hormones or neurotransmitters. When released from the adrenal glands they act as hormones. Norepinephrine and epinephrine act to keep the body vigilant and able to deal with a stressor by promoting alertness and increasing blood flow to skeletal muscles to ensure they are ready to act (Lovallo, 2016; McEwen, 2007)

The HPA-axis. The *Hypothalamic-Pituitary-Adrenal Axis* (HPA-Axis) also becomes activated when we come in contact with a stressor and is the most commonly studied physiological stress response system (Lovallo, 2016). Instead of sending a signal through the nervous system, the hypothalamus triggers a stress response through the release of hormones. When we experience something stressful, the hypothalamus releases a hormone called *Corticotropin-Releasing-Hormone* (CRH). CRH signals the *pituitary gland*, a hormone-secreting gland that is located inferior to the hypothalamus, to secrete a hormone called

Adrenocorticotrophic Hormone (ACTH) into the bloodstream (Lovallo, 2016). ACTH travels down to the adrenal glands where it prompts the release of a class of adrenal steroid hormones called glucocorticoids from the adrenal cortex, the outer layer of the adrenal glands. One of the glucocorticoids is cortisol, which plays an important role in the stress response (Lovallo, 2016).

The release of cortisol causes a number of changes that help the body to deal with stressors. For example, it helps to mobilize energy like glucose so the body has enough energy to cope with a prolonged stressor. When cortisol levels in the blood get high, this is sensed by receptors in areas of the brain like the hippocampus and leads to the shutting off of the stress response through what is known as a *negative feedback mechanism*. This mechanism is important in order to avoid overproduction of cortisol when it is no longer needed (Lovallo, 2016; McEwen, 2000).

Protecting and Damaging Effects of Stress Mediators - Allostasis and Allostatic Load

Allostasis is the adaptive process of maintaining homeostasis in face of a more chronic or severe stressor (McEwen, 2000). It is an active and energy consuming stress response that has to continue when the stressor is not eliminated. *Allostatic load* is the wear and tear on the body and brain that is caused by the costs of maintaining homeostasis through allostasis over a long period of time (McEwen, 2000). Systems in the body that are influenced by the stress response adapt to the stress in first hand, but a malfunctioning stress response causes damage to the systems (McEwen, 2000).

The immune system. The immune system is regulated by both the nervous and endocrine system. During acute stress, the body prepares itself for wounding. It is the immune defense system that is in charge of protecting the organism from infections and hence speeds up

the wound healing process during stressors (McEwen, 2000). Cells from the innate immune system enter the bloodstream, causing the number of immune cells in the blood to increase. These cells migrate to tissues in the body which are most likely to be damaged by the stressor. Once there, the immune cells are ready to aid healing by attacking microbes that may enter the body through tissue damage (Schneiderman et al., 2008).

However, in the face of a more chronic stressor, proinflammatory molecules produced by immune cells to attack viruses and bacteria become dysregulated. This is associated with slower wound healing and surgery recovery (Kiecolt-Glaser, Page, Marucha, MacCallum, & Glaser, 1998), higher vulnerability to infections, and poorer antibody responses to vaccination (Glaser, Sheridan, Malarkey, MacCallum, & Kiecolt-Glaser, 2000).

The cardiovascular system. When a stressor is detected, changes in the cardiovascular systems are triggered. The energy that is released during the activation of the stress response is distributed to the organs that need it the most through the blood flow. Hence the blood pressure increases to maintain the energy distribution (Schneiderman et al., 2008). This happens either by the contraction of certain blood vessels or through the increase of the heart rate and the amount of blood pumped out with each heartbeat. During stressful situations that do not require a lot of action, peripheral blood vessels contract to shunt the blood to internal organs. At the same time, this minimizes the potential bleeding in case of tissue damage. In situations that require more action, the heart rate and blood flow from the heart increases (Schneiderman et al., 2008).

The cardiovascular response can become maladaptive and cause cardiovascular diseases if it remains chronic, for example, sustained increases in blood pressure (Henry, Stephens, & Santisteban, 1975). It can also result in overactivation of the vascular stress response, by responding to all kinds of stressor even if not needed. These elevations can over time lead to

damage and plaque formation of the blood vessels which in turn may cause coronary heart disease, stroke, or other cardiovascular conditions (McEwen, 2007; Schneiderman et al., 2008).

The hippocampus. Several brain areas that are involved with mediating the stress response are themselves targets of the hormones released in the response, such as the hippocampus, amygdala, and prefrontal cortex (PFC). The hippocampus was the first brain region to be identified as a target of stress hormones (McEwen, 2007). As mentioned earlier, it is known that the hippocampus is an important brain structure regarding declarative, episodic and spatial learning and memory (McEwen, 1999). The hippocampus is also known to be a very plastic brain structure, meaning that it is easily changeable and modifiable. This makes it a very vulnerable structure that can be easily damaged by trauma as well as aging or stress (McEwen, 1999).

Adrenal steroids, such as cortisol, can act on two different types of receptors in the brain; Type 1 receptors, that are also called *mineralocorticoid receptors* and Type 2 receptors, *glucocorticoid receptors* (Lovallo, 2016). Both of these receptors are found in hippocampal neurons. Severe or repeated stress resulting in elevated adrenal steroids, such as glucocorticoids, has been correlated with atrophy of the human brain (Gurvits et al., 1996). Due to the fact that the hippocampus is such a sensitive brain area, it is particularly targeted and shows greater changes during severe stress than other brain areas (McEwen, 1999).

It is suggested that elevated adrenal steroids in the brain are associated with dendritic atrophy and suppressed neurogenesis (e.g. the birth of new neurons) in the hippocampus (Gould, McEwen, Tanapat, Galea, & Fuchs, 1997). This may cause cognitive dysfunctions such as impaired spatial and episodic memory, attention, and learning (McEwen, 1999). Individuals who have experienced severe stress show a significantly smaller hippocampus compared to less

impaired individuals (Gurvits et al., 1996). Furthermore, the hippocampus helps to shut off the HPA-axis when the stressor has been removed. A compromised hippocampus results in a prolonged HPA response, which leads to even more stress and wear and tear on the body (McEwen, 2007).

The amygdala and prefrontal cortex. The amygdala and PFC are also subjects to functional and structural changes due to acute and repeated stress. As the hippocampus, the PFC and amygdala contain adrenal steroid receptors (McEwen, 2007). However, the role of these mediators has not been studied in detail in these brain regions as it has in the hippocampus. Nonetheless, glucocorticoids may have a role in the dendritic shrinkage in medial PFC (mPFC) and loss of neurons in different areas of the PFC, such as the cingulate cortex (Cerqueira et al., 2005; Brown, Henning, & Wellman, 2005). These changes correlate with cognitive impairments in attention set-shifting and extinction of fear conditioning (Liston et al., 2006; Miracle, Brace, Huyck, Singler, & Wellman, 2006). While stress has shown dendritic shortening and neural loss in PFC, it is correlated with dendritic growth in the amygdala (Vyas, Mitra, Rao, & Chattarji, 2002). This leads to hyperactivity of the amygdala and hence enhance anxiety and amygdala-dependent unlearned fear and fear conditioning (Conrad, Magarinos, LeDoux, & McEwen, 1999).

Neurobiological Factors of Stress-Induced Psychopathology

Allostatic load is correlated with different kinds of psychopathology, such as MDD and chronic anxiety disorders like PTSD (McEwen, 2003). There are several neurobiological and psychosocial factors associated with stress-induced MDD and PTSD. Some neurobiological factors include dysregulation of serotonin, norepinephrine, the HPA-Axis, dopamine and the

reward system, galanin, estrogen, among others (Charney, 2004; Southwick, Vythilingam, & Charney, 2005). With consideration to the space limitation of this thesis, all factors will not be assessed. However, the first three will be explained further in the following paragraphs.

Serotonin. Chronic psychological stress has been correlated with a decrease of one kind of serotonin receptor, 5-HT1A, indicating an association with the release of CRH and cortisol during the stress response (López, Chalmers, Little, & Watson, 1998). Other factors contributing to the reduction could also be of a genetic character, or as a combination of genetic predisposition and psychosocial stress (Charney, 2004). Abnormal serotonergic function and reductions of serotonin receptors 5-HT1A have been strongly associated with stress-related MDD and abnormal regulation of anxiety behavior (Charney, 2004; Drevets, 2000; Gross et al., 2002)

Norepinephrine and neuropeptide Y. Many studies have shown support for norepinephrine to be an important feature in the pathophysiology of MDD. As noted earlier, epinephrine and norepinephrine are mediators of the stress response and released by the activation of the sympathetic nervous system (SNS). There are individual variations in the responsiveness of the SNS. A hyper-sensitive SNS that responds to minor stressors can lead to depression, fear, chronic anxiety, and cardiovascular diseases among others, due to the overproduction of norepinephrine (Southwick et al., 2005). Individuals with PTSD or MDD have shown this SNS hyper-responsiveness (Schnurr, & Green, 2004). Neuropeptide Y (NPY) is an amino acid that is released together with norepinephrine when the SNS is strongly activated. It works to regulate the release of norepinephrine to avoid overproduction (Southwick et al., 2005). A decrease of NPY can, therefore, result in elevated levels of norepinephrine. Decreased concentrations of NPY have been shown in individuals diagnosed with MDD and PTSD

compared to controls (Sah, Ekhtor, Jefferson-Wilson, Horn, & Geraciotti, 2014; Hou, Jia, Liu, & Li, 2006). Furthermore, low levels of NPY has also been shown in individuals who have attempted suicide recently (Westrin, Ekman, & Traskman-Bendz, 1999).

The HPA-axis. Patients with diagnosed MDD has consistently been shown to have alterations in HPA-axis functioning. Around half of the individuals diagnosed with MDD has a hyperactive HPA-axis. This hyperactivity can be mediated through the damage of the hippocampus that helps to regulate the HPA-axis. As mentioned previously, damage to the hippocampus can result in a dysregulation of the HPA-axis, since it is the hippocampus that helps to shut off the HPA-axis response. This damage can lead to hyperactivation of the HPA-axis. Additionally, antidepressants have been shown as having a normalizing function of the HPA-axis, bringing it back to a balanced state (Southwick et al., 2005).

So, overall, the research discussed regarding stress implies that there is a mediating effect between stress responses and maladaptive effects, such as physical and psychological pathologies. Given that chronic or severe stress can cause maladaptive changes in several systems in the body, it is of importance to not only seek the understanding of the mechanics of stress but also of possible interventions to build resilience and cope with stress (Johnson & Acabchuk, 2018). The following sections will assess these subjects, and begin with resilience.

Resilience and Vulnerability to Stress

Resilience to stress represents several forms of successful adaptations in both biological, cognitive, and behavioral responses in the face of traumatic events, and can therefore be difficult to operationalize (Osório, Probert, Jones, Young, & Robbins, 2016). Nonetheless, resilience has been defined as a dynamic process that promotes adaptation in the face of stressful and traumatic

events while physical and psychological functions remain relatively normal (Osório et al., 2016). A resilient individual is characterized by showing an advanced capacity to avoid damaging physical and psychological effects from severe stress, which otherwise would result in serious stress-related disorders, such as MDD and PTSD (Russo, Murrough, Han, Charney, & Nestler, 2012).

It is import to mention again that perceptions, interpretations, and responses to stress, and vulnerability or resilience to allostatic load and related pathologies, vary widely between individuals. These individual differences depend on several genetic, cognitive, developmental, neurobiological and psychological factors (Southwick & Charney, 2012). As discussed in the previous section, there are many neurobiological factors that contribute to the onset of stress-related psychopathologies, such as the regulation of the HPA-axis, serotonergic function, the release of norepinephrine and NPY, and many more which have not been assessed in the current thesis. It is these same factors that mediate individual resilience to stress. If neurobiological factors correlated vulnerability to stress and related diseases are a hyperactive SNS and HPA-axis, elevated concentrations of norepinephrine, and low levels of NPY and serotonin receptors, a resilient profile would be characterized by the opposite (Southwick & Charney, 2012).

Studies have also suggested that stressors during early life can be an important factor to consider when looking at individual vulnerability to stress-related psychopathology and stress resilience (Rende, 2012). Severe stress in early childhood has been associated with damage to the development of the stress response systems and can hence cause long-lasting consequences. Dysfunctional activation of the stress response systems in early life can, in turn, affect the normal development of the brain as well as other systems, and increase the risk of developing stress-related cognitive and emotional problems (Shonkoff & Garner, 2012). Studies have shown

that neglected babies show fewer stress management skills and higher levels of stress and anxiety. This was correlated with increased HPA-axis and CNS activity to stressors in later life. Exposure to early life stressors have also shown reductions in the hippocampus, and amygdala responsiveness to negative facial expressions (Dannlowski et al., 2012). Moreover, genetics play a critical role in the variance of resilience in different individuals, and hence vulnerability to stress-related psychiatric conditions (Osório et al., 2016). For example, twin-studies has shown that PTSD is 32-38% inheritable (True et al., 1993). In addition to these biological factors, there are many psychosocial factors that can contribute to building resilient individuals. One such factor is the use of different coping strategies that will now be further explained.

Coping

One approach to build stress resilience is through different coping strategies. Just like stress mediators, some coping strategies lead to adaptive outcomes and build resilience, while others are seen as maladaptive, leading to negative psychological outcomes. As noted earlier, the way in which an individual interprets or appraises a stressor is important in terms of the resultant response. When a situation has been interpreted as a challenge or threat during primary appraisal, the resultant distress elicits strategies to cope with the stressor, e.g. secondary appraisal. The cognitive process of deciding on a coping strategy includes the identification and evaluation of coping resources (e.g. self-efficacy), situational specific variables, and past experiences. These factors determine the coping actions aimed to change the relationship between the individual and the stressor (Biggs et al., 2017). After the actions aimed at coping with the stressor, a *cognitive reappraisal* process is triggered. The outcomes of the coping effort, together with new sensory information from the environment are again processed as in primary appraisal. This time we evaluate how well the coping actions have worked, and if the situation has changed. This

dynamic process between primary and secondary appraisal goes on until the situation is no longer perceived as a threat (Biggs et al., 2017).

Coping strategies can either be directed to handle the source stressor or to manage the emotions that arise due to the stressor. These different branches are classified as either *problem-focused coping*, where actions are aimed to change the source stressor, or *emotion-focused coping*, which is actions that aims to deal with the resultant emotional distress (Biggs et al., 2017). The effectiveness of these different strategies has been discussed within the research on coping. Most broadly, problem-focused coping has been associated with generally positive effects, such as improved psychological well-being, self-care, and health-related quality of life, and lower levels of emotional exhaustion. Conversely, emotion-focused coping has been regarded as generally maladaptive and ineffective, causing poorer psychological well-being and health-related quality of life, increased anxiety and dissatisfaction, and a higher risk of mortality (Mayordomo, Viguer, Sales, Satorres, & Meléndez, 2016). However, many researchers have argued against this generalization of the effects of problem-focused and emotion-focused coping. It has been suggested that neither coping strategy is essentially effective or ineffective, but that the effectiveness of any given coping strategy is dependent on the situation and appraisal of each specific stressful experience (Biggs et al., 2017). Emotion-focused coping has for example been shown to have short-term adaptive effects when the stressor generates great emotional distress, or when it is perceived as uncontrollable (Ben-Zur, 2009).

An alternative to the classification of coping strategies has been provided as *active* versus *passive* coping (Brown & Nicassio, 1987). Active coping strategies have been defined as the attempt to try to deal with the stressor by using own resources, and passive strategies are identified as helplessness or reliance on others. So the key distinction between these two is

whether the person relies on internal or external resources to control the stressor (Brown & Nicassio, 1987). For example, Active coping strategies have been linked to adaptive outcomes in chronic pain studies (Snow-Turek, Norris, & Tan, 1996). It has been shown to correlate to positive affect, better psychological adjustment, and a decrease in depression. Passive strategies have been correlated with maladaptive outcomes such as depression and increased pain (Snow-turek et al., 1996). Furthermore, depression has been correlated with passive, avoidant and emotional coping strategies in both community and clinical samples (Southwick et al., 2005). It has also been argued that coping style not only predicts the development of depression but that depression predicts use of coping strategies as well. In other words, it is a two-way process (Southwick et al., 2005)

Active coping, such as problem-focused coping, positive reappraisal, using humor, acceptance, seeking social support, and spirituality, has been argued to be a factor associated with resilience to stress and stress-induced depression (Mayordomo et al., 2016; Southwick et al., 2005). Different studies have linked resilient individuals with active coping, whereas low scores in resilience have been linked with alcohol abuse, behavioral and mental disengagement, and denial (Mayordomo et al., 2016; Southwick et al., 2005).

So, taken together, the following can be concluded regarding stress, stress-induced psychopathology, resilience, and coping. Looking at previous research, some coping strategies, such as denial and mental disengagement, have been linked to depression, anxiety, and distress (Deimling et al., 2006; Carver et al., 1993) whereas coping strategies such as acceptance, use of humor, and religious activity have been linked to stress resilience and lower distress (Carver et al., 1993; Carver, Scheier, & Weintraub, 1989). Building resilience to stress through adaptive coping strategies, among others, decreases the risk of developing stress-induced MDD and

PTSD. Based on these assumptions, the following predictions of the current study have been made.

Regarding the notion that Sri Lanka has the highest suicide rates in the world, and that psychopathology has been shown having a strong correlation to suicide, the first hypothesis predicts that Sri Lankans experience more symptoms of stress, anxiety, and depression compared to Swedes (H_1). Furthermore, it is hypothesized that Sri Lankans use maladaptive coping strategies to a greater extent than Swedes (H_2). The third hypothesis predicts that the use of maladaptive coping strategies should be correlated with higher levels of symptoms of stress, depression, and anxiety, whereas adaptive coping strategies should be correlated with lower levels (H_3).

Method

Design

Cultural differences in the use of different coping strategies and symptoms of stress, depression, and anxiety were studied through self-reported measurements in the form of two questionnaires. Data were collected in Sweden and Sri Lanka, forming two groups based on country. All variables were studied as between-group variables. Links between different coping strategies and levels of symptoms of stress, anxiety, and depression were studied as within-group variables in both groups.

Participants

A total of 145 individuals (Swedish, $n=77$, Sri Lankan, $n=68$) participated in the study. Due to omitting one of the two questionnaires, three participants were excluded from the

analyses (Swedish, $n=2$, Sri Lankan, $n=1$). Thus, data from a total number of 142 participants (Swedish, $n=75$, Sri Lankan, $n=67$) were further analyzed. Participants in Sweden were recruited by personal contact with the author by opportunity sampling. This mainly took place at the University of Skövde, but also in other random contexts. Participant age ranged from 18-49 years, with a mean age of 24.83 ($SD=5.4$). The gender distribution was 69.3% female ($n=52$) and 30.7% male ($n=23$). Sri Lankan participants were also recruited in contact with the author by opportunity sampling in schools, youth groups, and random contexts in Sri Lanka. The age of the participants ranged from 18-50 years, mean age being 23.63 ($SD=6.3$). There were 55.2% males ($n=37$), 43.3% females ($n=29$), and one participant failed to indicate sex (1.5%). No compensation was offered for participating in either Sweden or Sri Lanka.

Measurements

Demographic questions. Both Swedish and Sri Lankan participants were asked to indicate age and sex. In the Sri Lankan sample they were also asked how affected they were by a) the civil war in Sri Lanka, and b) the tsunami that affected Sri Lanka in 2004. They were instructed to indicate to what extent they have been affected on a Likert scale ranging from 1-4, 1 being *not at all* and 4 being *a lot*.

The Depression, Anxiety and Stress Scale (DASS). Symptoms of stress, anxiety, and depression were measured by the Depression, Anxiety and Stress Scale (DASS) (Lovibond & Lovibond, 1995). It consists of 42-items that are divided into three different subscales that independently measure the three different emotional states. The depression subscale measures emotional states as hopelessness, anhedonia, devaluation of life, lack of interest/involvement, dysphoria, self-deprecation, and inertia. The anxiety subscale treats situational anxiety,

subjective experience of anxious affect, skeletal muscle effects, and autonomic arousal. The stress subscale assesses nervous arousal, difficulty in relaxing, and being easily impatient, upset/agitated, and irritable/overactive. Each subscale consists of 14 items that are measured by indicating the severity/frequency they have experienced for each state over the past week on a Likert scale ranging from 1 - 4, 1 = *Did not apply to me at all*, 2 = *Applied to me to some degree, or some of the time*, 3 = *Applied to me a considerable degree, or a good part of the time*, and 4 = *Applied to me very much, or most of the time*. The sum of the scores for each subscale gives the total score for stress, anxiety, and depression. As mentioned, the original scale gave instructions to indicate the experience *over the past week*, however, as the author found this time frame to be too narrow and sensitive to change, this was changed, in accordance to instructions of the DASS, to *over the past month*.

The psychometrics of the DASS-scale have been studied within a large ($N=1771$) non-clinical western sample. Measured by Cronbach's alpha, the scale was shown to have good reliability, measuring .95 for depression, .90 for anxiety, .93 for stress, and .97 for the total scale (Crawford & Henry, 2003). The reliability of the shortened version of the DASS, with 21 items, has also been examined in eastern cultures. Adequate reliability has been shown in a sample of Vietnamese women, showing .88 for the overall scale (Tran, Tran, & Fisher, 2013), and .92 for the overall scale in a Chinese college student sample (Wang et al., 2016) Taken together, the DASS seems to be a reliable measurement in both western and eastern cultures.

The Brief COPE inventory. The Brief COPE inventory was used to measure different coping strategies used in relation to individual stressors (Carver, 1997). It is a shortened version of the COPE inventory and used in this study to shorten the length of the questionnaire to lessen the burden on the participants. The original scale consisted of 15 different coping strategies, with

4 item each (Carver et al., 1989). The Brief COPE is abbreviated and includes 14 different strategies of coping that are measured by 2 items each, making it a 28-item questionnaire. The different coping strategies measured are the following: *self-distraction* (e.g. mental disengagement), *active coping*, *denial*, *substance use*, *use of emotional support*, *use of instrumental support*, *behavioural disengagement*, *venting*, *positive reframing*, *planning*, *humor*, *acceptance*, *religion*, and *self-blame*. The participants are instructed to indicate to which extent they have been using each strategy on a scale from 1 - 4, where 1 = *I have not been doing this at all*, 2 = *I have been doing this a little bit*, 3 = *I have been doing this a medium amount*, and 4 = *I have been doing this a lot*. In this study, the coping strategies were divided into three different subscales based on the original article (Carver et al., 1989). *Problem-focused coping*, including active coping, instrumental support, and planning. *Emotion-focused coping*, including denial, emotional support, positive reframing, humor, acceptance, and religion. And the third category, *dysfunctional coping*, including self-distraction, substance use, behavioural disengagement, venting, and self-blame. In relation to the Brief COPE inventory, the participants of the current study were asked to define a stressor that had affected them recently and answer the subsequent questions based on actions aimed to cope with this defined stressor. The alternatives were the following: a) *Relationships*, b) *School*, c) *Money*, and d) *Other*.

The Brief COPE inventory has shown fairly good reliability in an American sample containing different ethnic groups. The Cronbach's alpha for each coping strategy ranged from .50 to .90, with all strategies except for acceptance, denial, and venting being over .64 (Carver, 1997). This making the scale adequately reliable. The validity and reliability of the inventory have also been studied in an Indian sample of HIV positive patients. The results showed

reasonable internal consistency, with alpha scores for each coping strategy ranging from .44 to .89, and a .70 alpha for the overall Brief COPE inventory (Mohanraj et al., 2015).

Procedure

All participants who wanted to take part in the study were first informed verbally about the study by the author. The participants were told that the study was within the field of cognitive neuroscience and about stress and health. They were informed that participation was anonymous and entirely voluntary, meaning that they were free to stop at any point. They were encouraged to answer as true to themselves as possible, and not to spend too much time on any item in the questionnaire. All participants were then given the opportunity to ask questions. More in-depth questions were answered after they had filled in the survey. Following the verbal instructions, all participants were handed an informed consent to read and sign before filling in the questionnaire (see Appendix A). All participants filled in the questionnaires by pen and paper. The questionnaire was designed with the DASS scale as the first part. The second part was coping, where an opportunity to define a current stressor was given before the Brief COPE inventory (see Appendix B). In the Sri Lankan sample, the questionnaire ended with two questions regarding the civil war and tsunami, as previously explained. There was no time limit for filling in the survey.

Sweden. The informed consent, DASS-scale and Brief COPE inventory were given in English in Sweden. This decision was made based on that Swedes are fairly good with the English language, and that the main context of data collection was among university students, assuming that this group could be even better at English than the average Swede. This was mentioned before the participants took part in the survey, and they were given the opportunity to

receive translation during the survey from the author if there were any trouble with understanding any word. The data were collected between January 19th-April 4th by opportunity sampling in Skövde, Stockholm, and Härnösand.

Sri Lanka. A translation of the informed consent, DASS-scale and Brief COPE inventory was made in Sri Lanka. Two independent translators performed backward translation from English to Sinhala (see Appendix C and D). Any mismatched words were corrected by a third party. Since English is a commonly used language in Sri Lanka, participants could choose which language they preferred when filling in the survey. When collecting data in groups that were not comfortable with English, a volunteer translated the verbal instructions by the author and possible questions by the participants. The data were collected between March 5th-March 30th by opportunity sampling in Negombo, Wattala, Kandy, and Weligama.

Results

All statistical analyses were carried out with IBM SPSS Statistics 24. Non-parametric statistical tests were applied in order to lessen effects of non-normal distribution in the sample. All analyses were two-tailed. There were very few missing responses, typically just one item missed by 8 respondents ($n_{SWE} = 3$, $n_{SRI} = 5$). In order to deal with these missing responses, the following computations were carried out. Where there was a missing response for an item in the DASS, the mean for that particular subscale for the participant was substituted. Where there was a missing response for the Brief COPE inventory, the same strategy was applied for the designated subscales (e. g. dysfunctional, problem-focused and emotion-focused coping). It was deemed that these adjustments would have a negligible effect on the overall analyses. As the DASS scaling was 1-4 in the current study, and in the original form was 0-3, comparisons with

normative data (see Discussion) required that the scores be adjusted by subtracting one point for each of the 14 items in each subscale in the discussion for that comparison. For the main statistics, the scaling remained at 1-4. Cronbach's alpha was used to measure the reliability of the two scales. The DASS had an alpha level of .86 for Sweden and .82 for Sri Lanka, indicating a good reliability. The Brief COPE inventory showed an alpha of .70 for Sweden and .77 Sri Lanka, also indicating a good reliability. No correction for multiple tests was applied.

Between-Group Differences

Analyzing the DASS subscales, the Sri Lankan sample showed statistically significant higher symptoms of stress ($Mdn = 32$, range = 30), anxiety ($Mdn = 27$, range = 32), and depression ($Mdn = 30$, range = 38) compared to the Swedish sample, which had $Mdn = 27$, range = 32 for stress ($n_{SWE} = 75$, $n_{SRI} = 67$, $U = 3321,5$, $p=.001$), $Mdn = 20$, range = 33 for anxiety ($n_{SWE} = 75$, $n_{SRI} = 67$, $U = 3622,5$, $p=.000$), and $Mdn = 21$, range = 37 for depression ($n_{SWE} = 75$, $n_{SRI} = 67$, $U = 3507,5$, $p=.000$). Results are shown in Table 1.

Table 1.

Mann-Whitney U differences in the median of total scores of Stress, Anxiety, and Depression between Sweden and Sri Lanka

Measure	<i>Mdn.</i> Sweden ($n=75$)	<i>Mdn.</i> Sri Lanka ($n=67$)	Sig. Level (two-tailed)
Total Stress Score	27	32	.001**
Total Anxiety Score	20	27	.000**
Total Depression Score	21	30	.000**

Note. *Mdn*=Median

Frequencies for the type of stressors in relation to used coping strategies are presented in Table 2. Many participants failed to define a stressor, and many participants filled in more than one option.

Table 2.

Frequencies for the type of stressors affected by participants in Sweden and Sri Lanka

Stressor	Sweden	Sri Lanka
Relationships	31	21
School	50	3
Money	15	16
Other	20	6

A Mann-Whitney U test was used to assess all the between-group differences. On average, the Sri Lankan sample used emotion-focused coping ($Mdn = 32$, range = 27) to a bigger extent than the Swedish sample ($Mdn = 28$, range = 28). These differences were statistically significant ($n_{SWE} = 75$, $n_{SRI} = 67$, $U = 3074$, $p = .022$). There were no other significant differences in coping style between the groups (see Table 3).

Table 3.

Mann-Whitney U differences in the median of total scores of Problem-focused, Emotion-focused and Dysfunctional coping between Sweden and Sri Lanka

Measure	<i>Mdn.</i> Sweden ($n=75$)	<i>Mdn.</i> Sri Lanka ($n=67$)	Sig. Level (two-tailed)
Problem-focused coping	17	17	.323
Emotion-focused coping	28	32	.022*
Dysfunctional coping	21	23	.795

Note. *Mdn*=Median

A few of the 14 different coping strategies showed statistically significant differences between the Swedish and Sri Lankan sample. Denial was more frequently used as a coping strategy in Sri Lanka ($Mdn = 5$, range = 6) compared to Sweden ($Mdn = 2$, range = 6) ($n_{SWE} = 75$, $n_{SRI} = 67$, $U = 3818$, $p = .000$). Behavioural disengagement was also more frequently used as a coping strategy in Sri Lanka ($Mdn = 4$, range = 5) compared to Sweden ($Mdn = 2$, range = 6) ($n_{SWE} = 75$, $n_{SRI} = 67$, $U = 3282$, $p = .001$). Furthermore, instrumental support was more used in the Sri Lankan sample ($Mdn = 5$, range = 6) compared to the Swedish sample ($Mdn = 4$, range = 6) ($n_{SWE} = 75$, $n_{SRI} = 67$, $U = 2998.5$, $p = .044$), as well as religion where $Mdn = 7$, range = 6 in Sri Lanka and $Mdn = 4$, range = 7 in Sweden ($n_{SWE} = 75$, $n_{SRI} = 67$, $U = 3697.5$, $p = .000$). Planning was the only coping strategy that was more frequently used in Sweden ($Mdn = 7$, range = 6) compared to Sri Lanka ($Mdn = 5$, range = 6) and was statistically significant ($n_{SWE} = 75$, $n_{SRI} = 67$, $U = 1899.5$, $p = .011$). These results are shown in Table 4. The rest of the coping strategies did not show any significant differences in frequency between the groups.

Table 4.

Significant Mann-Whitney U differences in the median of the 14 individual coping strategies between Sweden and Sri Lanka

Measure	<i>Mdn. Sweden</i> ($n=75$)	<i>Mdn. Sri Lanka</i> ($n=67$)	Sig. Level (two-tailed)
Denial	2	5	.000** ^a
Instrumental support	4	5	.044* ^a
Behavioral disengagement	2	4	.001** ^a
Religion	4	7	.000** ^a
Planning	7	5	.011* ^b

Note. Mdn=Median. ^a=Sweden<Sri Lanka. ^b=Sri Lanka<Sweden

Within-Group Correlations

Spearman's rho correlation was used to measure correlations between the DASS and the three categories of coping styles (e.g. dysfunctional, problem-focused, and emotion-focused coping) and all the 14 independent coping strategies included in the Brief COPE inventory in both the Swedish and Sri Lankan sample.

Sweden. The use of dysfunctional coping was on average significantly correlated with higher symptoms of stress ($\rho = .452, p < .01$), anxiety ($\rho = .591, p < .01$), and depression ($\rho = .640, p < .01$), as shown in Table 5. Problem-focused coping was negatively correlated with all subscales of the DASS in the Swedish sample, but were non-significant. The 14 individual coping strategies assessed in the Brief COPE were also analyzed for correlation with the DASS, showing more specific results. In the Swedish sample, denial, behavioural disengagement, mental disengagement, substance use, and self-blame were positively correlated with at least two of the three subscales of the DASS with a minimum significance level of $p < .05$ (see Table 7 for statistical details). Venting was positively correlated with stress ($\rho = .230, p < .05$), humor was surprisingly positively correlated with anxiety ($\rho = .284, p < .05$), and positive reframing was negatively correlated with depression ($\rho = -.237, p < .05$), also presented in Table 7.

Sri Lanka. The use of dysfunctional coping was on average significantly correlated with higher symptoms of stress ($\rho = .495, p < .01$), anxiety ($\rho = .374, p < .01$), and depression ($\rho = .340, p < .01$) in Sri Lanka as well. Furthermore, emotion-focused coping was significantly correlated with higher scores of stress ($\rho = .271, p < .05$), and noticeably, problem-focused coping was significantly correlated with a higher score of anxiety ($\rho = .259, p < .05$). These results are shown in Table 6. The Sri Lankan sample showed positive correlations between at least two of the three subscales included in the DASS and denial, self-blame, and humor, with a

minimum significance level of $p < .05$ (see Table 8 for statistical details). Mental disengagement and venting were positively correlated with stress ($p < .05$), and planning was perhaps surprisingly positively correlated with anxiety ($p < .05$), also shown in Table 8. There were no significant correlations between the DASS and being affected by the tsunami or civil war.

Table 5.

Spearman's rho correlations for Swedish sample of DASS and Problem-focused, Emotion-focused, and Dysfunctional coping

Measure	Total Stress Score	Total Anxiety Score	Total Depression Score
Problem-focused coping	-.023	-.039	-.099
Emotion-focused coping	-.015	.078	-.111
Dysfunctional coping	.452**	.591**	.640**

* Correlation is significant at the .05 level (two-tailed).

**Correlation is significant at the .01 level (two-tailed).

Table 6.

Spearman's rho correlations for Sri Lankan sample of DASS and Problem-focused, Emotion-focused, and Dysfunctional coping

Measure	Total Stress Score	Total Anxiety Score	Total Depression Score
Problem-focused coping	.169	.259*	.155
Emotion-focused coping	.271*	.183	.055
Dysfunctional coping	.495**	.374**	.340**

* Correlation is significant at the .05 level (two-tailed).

**Correlation is significant at the .01 level (two-tailed).

Within-Group Comparisons

In order to compare the extent of use of different coping styles within each group, an overall mean of dysfunctional, emotion-focused, and problem-focused was computed as a new variable. This took into consideration that each style consisted of a different number of items. A Wilcoxon Signed Rank Test was then carried out for both the Swedish and Sri Lankan samples. In the Swedish sample, problem-focused coping was on average more frequently used ($Mdn = 2.83$, range = 3) compared to both emotion-focused ($Mdn = 2.34$, range = 2.5, $Z = 163.5$, $p = .000$) and dysfunctional coping ($Mdn = 2.1$, range = 2.33, $Z = 178.2$, $p = .000$), and were statistically significant. Furthermore, it was also statistically significant that Swedes use emotion-focused coping more than dysfunctional coping ($Z = 181.8$, $p = .028$).

Similar findings were made in the Sri Lankan sample. Problem-focused coping was the most used coping style on average ($Mdn = 2.83$, range = 3) compared to both emotional-focused coping ($Mdn = 2.67$, range = 2.25, $Z = 128.84$, $p = .004$) and dysfunctional coping ($Mdn = 2.3$, range = 1.9, $Z = 160$, $p = .000$). Furthermore, emotion-focused coping was also significantly more used than dysfunctional coping ($Z = 156.53$, $p = .000$).

Table 7.*Spearman's rho correlations between the DASS and Brief COPE in Sweden*

	Measure	Total Stress Score	Total Anxiety Score	Total Depression Score
1.	Total Stress Score		.699**	.643**
2.	Total Anxiety Score			.629**
3.	Total Depression Score			
4.	Self distraction	.204	.376**	.319**
5.	Active coping	.089	-.014	-.130
6.	Denial	.291*	.315**	.339**
7.	Substance use	.182	.306**	.380**
8.	Emotional support	-.091	-.060	-.212
9.	Instrumental support	-.086	-.029	-.116
10.	Behavioural disengagement	.268*	.418**	.421**
11.	Venting	.230*	.147	.075
12.	Positive reframing	-.096	-.158	-.237*
13.	Planning	-.068	-.031	.030
14.	Humor	.118	.284*	.051
15.	Acceptance	-.189	-.077	-.061
16.	Religion	-.053	-.018	-.091
17.	Self-blame	.413**	.476**	.567**

* Correlation is significant at the .05 level (two-tailed).

**Correlation is significant at the .01 level (two-tailed).

Table 8.*Spearman's rho correlations between the DASS and Brief COPE in Sri Lanka*

	Measure	Total Stress Score	Total Anxiety Score	Total Depression Score
1.	Total Stress Score		.645**	.687**
2.	Total Anxiety Score			.600**
3.	Total Depression Score			
4.	Self distraction	.357**	.202	.191
5.	Active coping	.086	.105	.121
6.	Denial	.319**	.337**	.288*
7.	Substance use	-.084	.053	-.073
8.	Emotional support	.195	-.018	-.029
9.	Instrumental support	.07	.116	.028
10.	Behavioural disengagement	.206	.189	.236
11.	Venting	.286*	.139	.191
12.	Positive reframing	.163	.008	-.078
13.	Planning	.149	.259*	.201
14.	Humor	.309*	.337**	.175
15.	Acceptance	.103	.07	-.013
16.	Religion	-.003	.037	-.015
17.	Self-blame	.453**	.473**	.393**

* Correlation is significant at the .05 level (two-tailed).

**Correlation is significant at the .01 level (two-tailed).

Discussion

The first aim of this thesis was to provide a review of the main aspects of the cognitive neuroscientific mechanisms of stress and stress-induced pathology, as well as characteristics of stress resilience and coping strategies. The second aim stems from previous research concerning suicide, which has suggested that stress could be a critical risk factor. Severe life events causing acute or chronic stress, or stress-related psychopathologies, mainly characterized by depression and anxiety, are strongly linked to suicide in both western and eastern countries. Currently, studies concerning risk factors for suicide in Sri Lanka indicate that there is a difficulty of coping with severe life stressors. Based on this knowledge, the second aim was to investigate differences in use of coping strategies and symptoms of stress, anxiety, and depression in Sri Lanka compared to Sweden. Furthermore, a correlation between coping strategies and symptoms of stress, anxiety, and depression was of interest.

The most significant findings of this study suggest that 1) Sri Lankans experience more symptoms of stress, anxiety, and depression compared to Swedes, 2) dysfunctional coping is correlated with higher levels of stress, anxiety, and depression in both Sweden and Sri Lanka, 3) higher levels of stress predicts higher levels of anxiety and depression in both Sweden and Sri Lanka, and 4) both countries tend to favor problem-focused coping over emotion-focused and dysfunctional coping.

The first hypothesis stated that the Sri Lankan sample has higher scores of the DASS compared to the Swedish sample. The results support this hypothesis, with Sri Lankan participants having significantly more symptoms of all three subscales. It is however difficult to indicate why this difference exists. The means of the total score of the DASS subscales in both Sweden and Sri Lanka seems to be much higher compared to normative data. When designing

the scale, Lovibond and Lovibond (1995) collected normative data from a non-clinical Australian sample ($N=2914$). More than half of the participants were students ($n=1607$), which makes the norms accurate to this sample. The mean scores for that sample were: stress = 10.00 ($SD = 7.91$), anxiety = 4.7 ($SD = 4.91$), and depression = 6.34 ($SD = 6.97$) (Lovibond & Lovibond, 1995). In comparison with the Swedish sample, where stress = 13.97 ($SD = 7.1$), anxiety = 8.29 ($SD = 7.79$), depression = 10.56 ($SD = 10.47$), and the Sri Lankan sample, stress = 17.79 ($SD = 6.68$), anxiety = 13.4 ($SD = 7.24$), depression = 15.94 ($SD = 8.84$), it is clear that both Sweden and Sri Lanka have higher levels of all subscales. As mental illness and high stress are a general well-known phenomenon in Sweden, these results are not surprising. But for Sri Lanka, where mental illness has not been as studied, this finding could be of importance. With a total of 71 consultant psychiatrists in a country with a population of 20 million, it can be argued that Sri Lanka is a country where mental health care is not a priority (Medical Statistics Unit: Ministry of Health, Nutrition and Indigenous Medicine, 2017). As stated in the introduction, suicides in Sri Lanka seems to be mostly predicted by severe stressful life events. But as such events are not unique for Sri Lanka alone, it could be argued that there has to be another underlying factor.

As presented in the literature review, previous research has linked allostatic load to psychological impairments such as PTSD and MDD. Neurobiological factors including dysregulation of the HPA-axis resulting in hyperactivation of the SNS and an overproduction of for example glucocorticoids and norepinephrine, and a decrease in serotonin receptors and neuropeptide Y, have been correlated with abnormal regulation of anxiety behavior and depression (Charney, 2004; Southwick et al., 2005). These alterations have been found in individuals with stress related PTSD and MDD, and suicide victims (Sah et al., 2014; Westrin et

al., 1999). Since this study showed higher levels of stress, anxiety, and depression in the Sri Lankan Sample, it is suggested that these may be the mediators behind the link between severe life events and suicide attempts. The underreporting of mental illness in relation to suicide in Sri Lanka could be as a consequence of poor mental health care which in turn results in fewer diagnoses.

The second hypothesis in the current study predicted that Sri Lankans use maladaptive coping strategies more frequently than Swedes, which may partly be supported. Even though both countries tend to favor problem-focused coping, the findings of the current study suggest that Sri Lankans use emotion-focused coping to a bigger extent than Swedes. This goes in line with some previous research, that have indicated that individuals in eastern cultures are more prone to use emotion-focused coping compared to individuals in western cultures (Essau & Trommsdorff, 1996; Marsella, Escudero, & Gordon, 1972). It has previously been suggested that emotion-focused coping can predict negative outcomes such as poorer psychological well-being (Graven et al., 2014), increased anxiety, and emotional exhaustion (Boyd, Lewin, & Sager, 2009), but that it should not be deemed as fully maladaptive. Emotion-focused coping was however significantly correlated with higher levels of stress in the Sri Lankan sample. Furthermore, denial, which was categorized as emotion-focused coping, was correlated with higher levels of stress, anxiety, and depression in both countries. Humor, which also was included in emotion-focused coping surprisingly predicted higher scores of anxiety in both countries and also of stress in the Sri Lankan sample. Maladaptive effects of using denial as a coping strategy have been supported by previous research. In Deimling et al. (2006) they examined coping strategies in relation to anxiety, depression, and cancer-related worries in a sample of adult long-term cancer survivors. Their results suggested strong links between denial

and all three independent measures (Deimling et al., 2006). Furthermore, one study examining coping strategies and correlation to stress, anxiety, and depression in parents with children with autism found similar results. Denial predicted higher levels of both anxiety and depression (Hastings et al., 2005). Humor as a predictor of stress and anxiety, however, was not expected. Previous research has correlated use of humor with positive psychological outcomes, such as less stress (Kuiper, Martin, & Olinger, 1993), and lower symptoms of anxiety and depression (Litman, 2006). However, the use of humor as a coping strategy in India has been considered as inappropriate, which also may apply in the Sri Lankan culture (Mohanraj et al., 2015). Hence, cultural differences should be considered as a factor.

As some aspects of emotion-focused coping can be considered as maladaptive in this study, it could also be argued that these results show support for the first hypothesis. The results of the current study also give support for the notion that Sri Lankans use denial and behavioural disengagement more than Swedes which are related to negative outcomes (Carver et al., 1993; Deimling et al., 2006). However, the Sri Lankan sample had a higher frequency of some adaptive coping strategies as well, such as instrumental support and religion. Therefore it can be implied that Sri Lankans use maladaptive coping strategies to a bigger extent than Swedes and hence supports the first hypothesis, but it is by importance to note that they also use some adaptive coping strategies more frequently than Swedes as well.

One factor that may have an effect on this outcome could be the operational definition of emotion-focused coping. Denial, which in this study was included in emotion-focused coping could also be labeled as a dysfunctional coping strategy considering the negative aspects that are linked to it. Furthermore, all coping strategies classified as dysfunctional coping in this study could be argued as essentially emotion-focused since these strategies too are aimed at handling

the emotions that arise due to the stressor. If, for example, denial would have been included in dysfunctional coping instead, or that dysfunctional coping would be a part of emotion-focused coping when computing the data, the results presented in this study would most possibly be different. In Boyd et al. (2009) as cited earlier, this was the case. They measured coping from the two categories emotion- and problem-focused coping and applied the scale designed in Latack (1986) which consists of three subscales of coping: *control*, *escape*, and *symptom management*. Control was categorized as problem-focused coping and escape and symptom management were included in emotion-focused coping (Boyd et al., 2009). Hence, the emotion-focused coping scale included items that in the current study were categorized as dysfunctional coping, such as substance use, self-distraction, and behavioural disengagement. This, of course, affected their results, that emotion-focused coping was associated with increased anxiety and emotional exhaustion. It is therefore difficult to compare the current results with previous studies since the operational definition of adaptive/maladaptive and emotion/problem-focused coping are not the same. If the items included in the dysfunctional subscale of the Brief COPE would be categorized as emotion-focused instead, a stronger correlation would be shown between emotion-focused coping and the DASS. This would give a stronger support for the first hypothesis, that Sri Lankans use maladaptive coping strategies to a bigger extent than Swedes. Furthermore, which coping strategies that are correlated with adaptive or maladaptive outcomes seems to vary cross-culturally. It is therefore important to carefully consider the operational definitions and cultural context before drawing any conclusions about the correlation between coping strategies and other variables, and also when comparing such analyses with other studies.

As regarding the third hypothesis, dysfunctional coping predicted higher rates of all DASS subscales in both countries, and hence supports it. However, there were no significant

findings that problem- or emotion-focused coping predicted lower scores of the DASS. Nonetheless, in the Swedish sample, problem- and emotion-focused coping were negatively correlated with almost all DASS subscales even though not statistically significant. In the Sri Lankan sample, however, except for emotion-focused coping being associated with higher stress as previously discussed, problem-focused coping was positively correlated with higher anxiety. This could be mediated by the correlation between planning and anxiety, as shown in the Sri Lankan sample. These findings are quite surprising and could be approached from a biopsychosocial perspective. As mentioned previously in the thesis, it cannot be concluded that specific coping strategies can be fully adaptive or maladaptive, but that individual and situational characteristics determine which aspects of the strategy that is adaptive or maladaptive. The current finding supports this notion, and could possibly be mediated by a cultural difference, or just individual variations. As the samples of the current study were not particularly large and possibly not fully representative of the normal population in either Sweden or Sri Lanka, it is difficult to draw any conclusions to the general population based on these findings.

Another interesting finding of the study was that higher levels of stress were correlated with higher levels of anxiety and depression in both countries. This supports previous research regarding maladaptive consequences of stress, including stress-induced psychopathology. The results indicated, however, a mediating effect between all the three subscales. As there was a correlation between all three scales, it is difficult to draw any causal relationships.

Furthermore, looking at the frequencies of the defined stressors for both countries it can be noted that the main sources of stress differ. As the Swedish sample reported the most common source to be school, this was not the case in the Sri Lankan sample, even though more than half of the sample were students. What seems to be the leading causes of stress in the Sri Lankan

sample were relationships and money. These results are however very tentative regarding the fact that many participants failed to fill in this item, and that many participants filled in more than one option. However, it could be argued that school may not be a priority to stress about in Sri Lanka, but that other things may be more important and hence results in greater related stress.

The current study includes several limitations. Regarding the theoretical overview, there are not many studies yet conducted concerning cultural differences in resilience and coping. This makes it difficult to interpret the results of the empirical investigation. Since most of the research conducted within this area are based on western studies, the results from the Swedish sample were mostly supported by previous research. The Sri Lankan sample differed however in some ways from the existing literature on coping and the relationship to stress, anxiety, and depression.

This difference could also possibly be mediated by limitations of the data collection. Since opportunity sampling was used to recruit participants, there were some differences in the samples. Even though there were many students in both samples, it was only university students in the Swedish sample and high school students in the Sri Lankan sample. This difference can interfere with the results regarding different age groups, economic standards, level of education, and so forth. Being that the sample sizes were fairly small and that the mean age of both countries being approximately 24 years, they are also not very representative of the general population. Further, the Swedish sample had an overrepresentation of females, and the Sri Lankan sample was two third males, which may also affect the results. Besides limitations of the samples, differences in the translations of the surveys may have an effect. Since the translations were not made with regard to cultural differences, there may be items that could be viewed very differently depending on cultural contexts.

Future studies should continue to investigate cultural differences in stress resilience and coping from a neurobiological and psychosocial approach. Furthermore, more studies are needed concerning the high suicide rates in Sri Lanka. It is still unclear why the problem is so widely spread, and this study can have provided some indication regarding the high symptoms of stress, anxiety, and depression. A replication of the current study within a larger sample more representative for the general population in Sri Lanka could possibly give more robust findings. Furthermore, it would have been of interest to include questionnaires about resilience and suicide ideation in order to get a broader view of the problem.

Taken together, the following conclusions regarding the current thesis can be stated. There is some neurobiological evidence connecting severe or chronic stress to psychopathologies and suicide. The main mediators seem to be dysregulation of stress response systems, such as the HPA-axis and overactivation of the SNS resulting in overproduction of norepinephrine, glucocorticoids, among others. These neurobiological characteristics have been shown in individuals diagnosed with MDD and PTSD. The same elements appear to mediate the neurobiological factors of resilience to stress. Further, some coping strategies have been shown to build resilience to stress, as others are linked to maladaptive outcomes. The findings of the current study imply that Sri Lankans have more symptoms of stress, anxiety, and depression compared to Swedes. It is, however, unclear why this difference exists since there were no strong results concerning differences in use of poor coping strategies between the countries. The results also showed that both countries tend to favor problem-focused coping comparing with emotion-focused and dysfunctional coping. Implications of these findings could be to improve mental health care in Sri Lanka. Future studies should continue to try to investigate the underlying

factors of the high suicides in Sri Lanka, and also continue the research of cross-cultural differences in suicide risk factors, stress resilience, and coping strategies.

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Appendix A – English Version of the Informed Consent

INFORMED CONSENT

You are invited to take part in a survey which is being carried out as part of a field study to gather data for a thesis that will be written in the area of cognitive neuroscience.

Participation is entirely voluntary and you are free to stop at any point.

The topic of interest is stress and health. The study will be explained to you briefly and potential questions will be answered after you have filled in the questionnaires. It will take around 10 minutes to fill in the survey.

All data collected will be dealt with according to current Swedish rules relating to research data to prevent the violation of personal integrity in the processing of personal data. Analysis will be anonymous and personal integrity protected. Data will be stored separately from other materials concerning the study and destroyed at the end of the course.

Your invested time and participation is much appreciated. Thank you.

DECLARATION

The meaning and implications of taking part in the study have been explained verbally to me. I have read the written information. I have also been given the opportunity to ask questions to the researcher and have received answers to them.

☐ I hereby sign the declaration and agree to participate in the study

Place and date: _____

Signature: _____

Name of the researcher: Theresa Skaf

Name of the supervisor: Judith Annett

Department of Cognitive Neuroscience & Philosophy
University of Skövde
Judith.annett@his.se

Appendix B – English Version of the Questionnaire

1. Age

2. Gender

☐ Male

☐ Female

3. Participant no

Part 1: Stress

The following statements describe symptoms of stress. Please read each statement and circle a number 1, 2, 3 or 4, which indicates how much the statement applied to you ***over the past month***. There are no right or wrong answers. Please answer as true for you as possible. Do not spend too much time on any statement.

The rating scale is as follows:

1 = Did not apply to me at all

2 = Applied to me to some degree, or some of the time

3 = Applied to me to a considerable degree, or a good part of time

4 = Applied to me very much, or most of the time

1. I found myself getting upset by quite trivial things	1	2	3	4
2. I was aware of dryness of my mouth	1	2	3	4
3. I couldn't seem to experience any positive feeling at all	1	2	3	4
4. I experienced breathing difficulty (eg, excessively rapid breathing, breathlessness in the absence of physical exertion)	1	2	3	4
5. I just couldn't seem to get going	1	2	3	4
6. I tended to over-react to situations	1	2	3	4
7. I had a feeling of shakiness (eg, legs going to give way)	1	2	3	4
8. I found it difficult to relax	1	2	3	4
9. I found myself in situations that made me so anxious I was most relieved when they ended	1	2	3	4
10. I felt that I had nothing to look forward to	1	2	3	4
11. I found myself getting upset rather easily	1	2	3	4
12. I felt that I was experiencing a lot of nervous energy	1	2	3	4
13. I felt sad and depressed	1	2	3	4
14. I found myself getting impatient when I was delayed in any way (eg, lifts, traffic lights, being kept waiting)	1	2	3	4

15. I had a feeling of faintness	1	2	3	4
16. I felt that I had lost interest in just about everything	1	2	3	4
17. I felt I wasn't worth much as a person	1	2	3	4
18. I felt that I was rather sensitive	1	2	3	4
19. I sweated noticeably (eg, hands sweaty) in the absence of high temperatures or physical exertion	1	2	3	4
20. I felt scared without any good reason	1	2	3	4
21. I felt that life wasn't worthwhile	1	2	3	4
22. I found it hard to wind down	1	2	3	4
23. I had difficulty in swallowing	1	2	3	4
24. I couldn't seem to get any enjoyment out of the things I did	1	2	3	4
25. I was aware of the action of my heart in the absence of physical exertion (eg, sense of heart rate increase, heart missing a beat)	1	2	3	4
26. I felt down-hearted and sad	1	2	3	4
27. I found that I was very irritable	1	2	3	4
28. I felt I was close to panic	1	2	3	4
29. I found it hard to calm down after something upset me	1	2	3	4
30. I feared that I would be "thrown" by some trivial but unfamiliar task	1	2	3	4
31. I was unable to become enthusiastic about anything	1	2	3	4
32. I found it difficult to tolerate interruptions to what I was doing	1	2	3	4
33. I was in a state of nervous tension	1	2	3	4
34. I felt I was pretty worthless	1	2	3	4
35. I was intolerant of anything that kept me from getting on with what I was doing	1	2	3	4
36. I felt terrified	1	2	3	4
37. I could see nothing in the future to be hopeful about	1	2	3	4
38. I felt that life was meaningless	1	2	3	4
39. I found myself getting disturbed	1	2	3	4
40. I was worried about situations in which I might panic and make a fool of myself	1	2	3	4
41. I experienced trembling (eg, in the hands)	1	2	3	4
42. I found it difficult to work up the initiative to do things	1	2	3	4

Part 2: Coping

These items deal with ways you've been coping with stressors in your life. I want you to think about a specific stressor that has affected you recently and define it below. If you don't have any current stressor, you can answer by writing "nothing". Remember that all data collection is anonymous and nothing can be traced back to you.

The stressor that has affected me is related to:

☐ Relationships ☐ School ☐ Money ☐ Other: _____

There are many ways to try to deal with problems. I want to know to what extent you've been doing what each item below says, to deal with your defined stressor. How much or how frequently. Don't answer on the basis of whether it seems to be working or not—just whether or not you're doing it. Try to rate each item separately in your mind from the others. Make your answers as true FOR YOU as you can. If you don't have any current stressor, answer by what you think you would have done if you had a stressor that affected you. Do not spend too much time on any statement.

The rating scale is as follows:

- 1 = I haven't been doing this at all
- 2 = I've been doing this a little bit
- 3 = I've been doing this a medium amount
- 4 = I've been doing this a lot

1. I've been turning to work or other activities to take my mind off things	1	2	3	4
2. I've been concentrating my efforts on doing something about the situation I'm in	1	2	3	4
3. I've been saying to myself "this isn't real"	1	2	3	4
4. I've been using alcohol or other drugs to make myself feel better	1	2	3	4
5. I've been getting emotional support from others	1	2	3	4
6. I've been giving up trying to deal with it	1	2	3	4
7. I've been taking action to try to make the situation better	1	2	3	4
8. I've been refusing to believe that it has happened	1	2	3	4
9. I've been talking about my unpleasant feelings with someone	1	2	3	4
10. I've been getting help and advice from other people	1	2	3	4

11. I've been using alcohol or other drugs to help me through it	1	2	3	4
12. I've been trying to see it in a different light, to make it seem more positive	1	2	3	4
13. I've been criticizing myself	1	2	3	4
14. I've been trying to come up with a strategy about what to do	1	2	3	4
15. I've been getting comfort and understanding from someone	1	2	3	4
16. I've been giving up the attempt to cope	1	2	3	4
17. I've been looking for something good in what is happening	1	2	3	4
18. I've been making jokes about it	1	2	3	4
19. I've been doing something to think about it less, such as going to movies, watching TV, reading, daydreaming, sleeping or shopping	1	2	3	4
20. I've been accepting the reality of the fact that it has happened	1	2	3	4
21. I've been expressing my negative feelings	1	2	3	4
22. I've been trying to find comfort in my religion or spiritual beliefs	1	2	3	4
23. I've been trying to get advice or help from other people about what to do	1	2	3	4
24. I've been learning to live with it	1	2	3	4
25. I've been thinking hard about what steps to take	1	2	3	4
26. I've been blaming myself for things that happened	1	2	3	4
27. I've been praying or meditating	1	2	3	4
28. I've been making fun of the situation	1	2	3	4

Part 3: Cultural Specific Circumstances

The rating scale is as follows:

1 = Not at all

2 = A little bit

3 = A medium amount

4 = A lot

1. How affected were you by the civil war in Sri Lanka?	1	2	3	4
2. How affected were you by the tsunami in 2004?	1	2	3	4

Appendix C – Sinhala Version of the Informed Consent

තොරතුරු අනුමතිය

ප්‍රජනන ස්නායු විද්‍යා ක්ෂේත්‍රයට ආශ්‍රිතව අධ්‍යයන කර දුන්න සොයා ගැනීම සඳහා සිදු කරනු ලබන සමීක්ෂණයක් සඳහා සහභාගි වීමට ඔබට ආරාධනා කරනු ලැබේ.
සහභාගි වීම මුලුමනින්ම ස්වේච්ඡාවෙන් සිදු වන අතර ඕනෑම ස්ථානයක නතර කිරීමට ඔබට නිදහස තිබේ.

උනන්දුව දක්වන මාතෘකාව වන්නේ ආතතිය හා ශාරීරික සුව නාවයයි . මෙම අධ්‍යයනය ඔබට කෙටියෙන් විස්තර කර ඔබ ප්‍රශ්නාවලිය පුරවා අවසන් වූ පසු විභව ප්‍රශ්නාවලට පිළිතුරු ලබා දෙනු ඇත. මෙම සමීක්ෂණය පිරවීම සඳහා විනාඩි 10 ක් ගතවනු ඇත.

සියලුම පුද්ගලික දත්ත සැකසීමේ දී පුද්ගලික අඛණ්ඩතාව උල්ලංඝනය කිරීම වැළැක්වීම සඳහා පර්යේෂිත දත්ත වත්මන් ස්විඩන නීතිවලට වලට අදාළව ගනන් ගනු ලැබේ. එකතු කරන ලද සියලු පුද්ගලික දත්ත විශ්ලේෂණය නිර්නාමිකව සිදුකරන අතර එහි අඛණ්ඩතාව ආරක්ෂා වනු ඇත. පාඨමාලාවේ අවසානයේ දී අධ්‍යයනයට ගෙන ගබඩා කළ දත්ත සියල්ල විනාශ කෙරේ.
ඔබේ ආයෝජනය කරන කාලය හා සහභාගිත්වය අගය කරනු ලැබේ.
ඔබට ස්තූතියි.

ප්‍රකාශනය

මෙම අධ්‍යයනයට හවුල් වීමෙහි අර්ථය හා ප්‍රතිවිපාක වාචිකව මා වෙත පැහැදිලි කර ඇත.
මම මෙහි ලිඛිත තොරතුරු කියවා ඇත. ප්‍රශ්න අසා පර්යේෂකයාගෙන් පිළිතුරු ලබා ගැනීමේ අවස්ථාව මා හට ලැබී ඇත.



මෙම ප්‍රකාශය අත්සන් කොට මෙම අධ්‍යයනයට සහභාගි වීමට එකඟ වෙමි.

ස්ථානය සහ දිනය :

අත්සන :

පර්යේෂකයාගේ නම : තෙරේසා ස්කාට්

අධීක්ෂකගේ නම : ජුඩිත් ඇනට්

ප්‍රජනන ස්වභාවික විද්‍යා හා දර්ශන විද්‍යා දෙපාර්තමේන්තුව

Skovde විශ්ව විද්‍යාලය

Judith.annett@his.se

Appendix D – Sinhala Version of the Questionnaire

1. වයස

2. ලිංගභේදය

3. Participant no

○ පුරුෂ

○ ස්ත්‍රී

1 වන කොටස : ආතතිය

පහත සඳහන් ප්‍රකාශ ආතතියේ ලක්ෂණ විස්තර කරයි. කරුණාකර සෑම ප්‍රකාශයක්ම කියවා පසු ගිය මාසය පුරාවට ඔබට ප්‍රකාශය කෙතරම් ආදාළ වී ඇති 1,2,3 හෝ 4 රටුම් කරන්න. නිවැරදි හෝ වැරදි පිළිතුරු නොමැත. කරුණාකර ඔබට හැකි පරිදි සත්‍ය පිළිතුරු දෙන්න. ඕනෑම ප්‍රකාශයක් මත වැඩි කාලයක් ගත නොකරන්න.

ණය ශ්‍රේණිය පහත පරිදි වේ.

1 = මට කිසිසේත්ම අදාළ නොවීය.

2 = කිසියම් ප්‍රමාණයකට හෝ යම් කාලයක් සඳහා මා හට අදාළ විය.

3 = මට සැලකිය යුතු තරමට, හෝ තරමක වේලාවක් මා හට අදාළ විය.

4 = මට බොහෝ වේලාවට හෝ වැඩි කාලයක් තුළදී අදාළ විය.

1. මම තරමක් නොවැදගත් දේවල් නිසා කණස්සල්ලට පත් වුණා	1	2	3	4
2. මගේ මුඛයේ වියළි බව මා ඇත සිටියෙමි.	1	2	3	4
3. මට කිසිදු ධනාත්මක හැඟීමක් ඇති බව පෙනෙන්නට නොහැකි විය.	1	2	3	4
4. මම උසුම ගැනීමේ දුෂ්කරතාවයක ද (උදා: අධික වේගයෙන් හුස්ම ගැනීම, ශාරීරික ක්‍රියාකාරීත්වය නොමැතිව හුස්ම නිර වීම.)	1	2	3	4
5. මට ඉදිරියට යන්න හැකි බවක් පෙනෙන්නට නිබුණේ නැත.	1	2	3	4
6. මම අවස්ථා ගණනාවකදී අධි ප්‍රවීණතාවයකට පත් විය.	1	2	3	4
7. මට වේවිලන ගතියක් ඇතිවේ. (උදා: කකුල් සෙලවෙන ස්වභාවයක්)	1	2	3	4
8. මට විවේකීව සිටීමට නොහැකි විය.	1	2	3	4
9. මා බොහෝ කණස්සල්ලට පත් වූ අවස්ථාවකදී මම සොයා ගන්නා එය අවසන් වූ පසු මට ඉතාම සැතපීමක් ලැබුණු බව.	1	2	3	4
10. ඉදිරිය ගැන සිතීමට කිසිම දෙයක් මට හැකි බව හැඟුණා.	1	2	3	4
11. මම තරමක් ඉක්මණින් අසහනයට පත් වෙමි.	1	2	3	4
12. මට හැඟුණේ ස්නායු ශක්තියෙන් ගොඩක් අත්දැකීම් මම ලබන බවයි.	1	2	3	4
13. මම දුකට හා කලකිරීමට පත් වුණා.	1	2	3	4
14. මම ප්‍රමාද වී සිටිය අවස්ථාවකදී මම නොඉවසිලිමත් වීම (උදා: විදුලි රැහැන්, වාහන ආලෝකය, බලා සිටීම)	1	2	3	4

15. මට ක්ලාන්ත හැඟීමක් දැණුනා.	1	2	3	4
16. මට හැම දෙයක් ගැනම උනන්දුවක් නැති වේගෙන එන බව දැණුනා.	1	2	3	4
17. මම පුද්ගලයෙක් හැටියට මට මාව වටින්නේ නැහැ කියලා හැඟුණා.	1	2	3	4
18. මට සංවේදී බවක් දැණුනා.	1	2	3	4
19. අධික උෂ්ණත්වයක් හෝ කායික පීඩනයක් නොමැතත් අධික දහඩිය වහනයක් සිදුවේ (උදා:අත් වල)	1	2	3	4
20. මම කිසිම හේතුවක් නැතුව බය වුණා.	1	2	3	4
21. ජීවිතය වටින්නේ නැති බව මට හැඟුණා.	1	2	3	4
22. මට වැටහුණා යමක් වටහා ගැනීමට අමාරු බවක්.	1	2	3	4
23. මට ගිලිමේ අපහසුතාවයක් ඇති විය.	1	2	3	4
24. මා කළ දේවල් වලින් සතුටක් ලැබීමට මට නොහැකි විය.	1	2	3	4
25. ශරීරික ක්‍රියාකාරීත්වය නොමැතිව,මගේ හදවතේ ක්‍රියාකරීත්වය ගැන මම දැන සිටියෙමි.	1	2	3	4
26. මට දුකක් හා සිතින් වැටී ඇති බවක් දැණුනා.	1	2	3	4
27. මම ඉතා කුපිත වු බවක් මට වැටහුණා.	1	2	3	4
28. හදිසි මහත් බයක් ඇති වීමට ආසන්න බවක් මට දැණුනා.	1	2	3	4
29. මට මොකක් හරි අවුලක් වුණාම සන්සුන් වෙන්න මට අමාරු වුණා.	1	2	3	4
30. සමහර නොවැදගත් දේවල් විසින් මාව “අහක් කරයි” කියා බය වුණු නමුත් නුතරු කටයුත්තකි.	1	2	3	4
31. ඕනෑම දෙයක් ගැන උද්යෝගිමත් වීමට මට නොහැකි විය.	1	2	3	4
32. මා කළ දෙයට බාධා ඉවසා සිටීම මට අමාරු විය.	1	2	3	4
33. මම සිටියේ සුනායු ආතතියකින්.	1	2	3	4
34. මට හැඟුණා මම තරමක් නොවටිනා බව.	1	2	3	4
35. මා කරමින් සිටි දේ සමඟ නොපැකිලෙන ඕනෑම දේයකට මම අකමැති විය.	1	2	3	4
36. මම තීතියට පත් වුණා.	1	2	3	4
37. මට කිසිවක් අනාගතය ගැන බලාපොරොත්තුවක් නැති වුණා.	1	2	3	4
38. ජීවිතය මට තේරුමක් නැති බව මට හැඟුණා.	1	2	3	4
39. මට මාවම වදයක් ලෙස හැඟුණා.	1	2	3	4
40. මම මම ගැනම කම්පනයට හෝ මෝඩයෙකු බවට පත් වෙයි කියා සිතා යම් අවස්ථාවන් වල මම කණාස්සල්ලට පත් වුණා.	1	2	3	4
41. මා වෙවිලීම අත් දුටුවා. (උදා:අත් වල)	1	2	3	4
42. යම් දෙයක් කිරීමට මුල පිරීම මට අමාරු වැඩක් වුණා.	1	2	3	4

2 වන කොටස: ගැටීම.

මෙම අයිතමයන් ඔබේ ජීවිතයේ පීඩකයන් සමග ගැටෙන ආකාරයන් ගැන වේ. මෑතකදී ඔබට බලපෑමක් ඇති කල ආතතික පීඩකයෙක් ගැන ඔබ සිතා බලා එය පහත දැක්වෙන ලෙසත් අර්ථකථනය කරන්න. ඔබට දැනට ආතතික පීඩකයෙක් නොමැති නම් ඔබට "කිසිවක් හැට" ලිවීමෙන් පිළිතුරු දිය හැකිය. සියලු දත්ත එක්රැස් කිරීම නිර්නාමික බව මතක තබාගන්න. කාටත් කිසිවක් සොයා ගත නොහැක.

මට බලපෑම් කළ මට බලපෑ ආතතික පීඩකයා සම්බන්ධ වන්නේ:

0 සබඳතා

0 පාසැල්

0 මුදල්

0 වෙනත්: _____

ප්‍රශ්න සමඟ කටයුතු කිරීමට උත්සාහ දැරීමේ බොහෝ ක්‍රම තිබේ. පීඩනය ඇතිකරන්නා පිළිබඳව ඔබ කුමන මට්ටමකට ද පහත සඳහන් එක් එක් අයිතමය සමඟ කටයුතු කරමින් සිට ඇත්තේ කියා මා දැන ගත යුතු වේ. කොපමණ හෝ කොපමණ වාර ගණනක් එය වැඩ කරනවා ද නැද්ද යන පදනම මත නොවැඩිවීම ඔබ එය කරනවා ද නැද්ද යන්න මත පමණක් පිළිතුරු දෙන්න. අනෙක් අයිතමයන් ඔබේ මනස තුළ වෙන වෙනම ශ්‍රේණි ගතකිරීමට උත්සාහ කරන්න. ඔබේ පිළිතුර හැකිතරම් ඔබටම සැබෑවක් වන අන්දමට සපයන්න. වර්තමානයේ ඔබට එවැනි පීඩකයෙක් නැතිනම් ඔබට එසේ බලපෑ පීඩකයෙක් තිබුනේ නම් ඔබ කුමක් කරයි ද යනුවෙන් පිළිතුරක් දෙන්න. ඕනෑම ප්‍රකාශයක් මත වැඩි කාලයක් ගත නොකරන්න.

ණය ශ්‍රේණිය පහත පරිදි වේ:

1 =මා මේ සියල්ල කරන්නේ නැහැ

2 =මා මෙය ටිකක් කර ඇත

3 =මා මෙය මධ්‍යම ප්‍රමාණයෙන් කර ඇත

4 =මා මෙය බොහෝසෙයින් කර ඇත

1. මගේ සිතින් යම් කාරණා මග හරවා ගැනීමට මම විවිධ වැඩ කටයුතුවල නිරත වුණා .	1	2	3	4
2. මම මගේ තත්වය පිළිබඳව යමක් කිරීමට මගේ උත්සහය අවධානය යොමු කර තිබේ.	1	2	3	4
3. "මෙය සත්‍යක් නොවේ" යැයි මම මටම කියා ගනිමි.	1	2	3	4
4. මම මාවම සහසා ගැනීමට මත්පැන් හෝ මධ්‍යසාර භාවිතා කරමින් සිටියා.	1	2	3	4
5. මම අන් අයගෙන් විත්ත වේගිය සහය ලබා ගත්තා.	1	2	3	4
6. මම එය සමඟ ගණුදෙනු කිරීමට දරණ උත්සහය අත් හැරියා.	1	2	3	4
7. මම එම තත්වය යහපත් කිරීමට උත්සහ දැරුවා.	1	2	3	4
8. එය සිදු වී ඇති බව විශ්වාස කිරීමට මා ප්‍රතික්ෂේප කළා.	1	2	3	4
9. මම කෙනෙකු සමඟ මගේ අප්‍රසන්න හැඟීම් ගැන කතා කරමි.	1	2	3	4

10. මම වෙනත් අයගෙන් උදව් සහ උපදෙස් ලබා ගනිමින් සිටියා.	1	2	3	4
11. මම එය හරහා මට උදව් කිරීමට මත්පැන් හෝ වෙනත් මධ්‍යසාර භාවිතා කරමින් සිටියා.	1	2	3	4
12. මම එය තවත් ආලෝකය තුළ දැකීමට උත්සහා කර ඇත,එය වඩා ධනාත්මක ලෙස දැකීමට	1	2	3	4
13. මම මාවම විවේචනය කරමින් සිටියා.	1	2	3	4
14. මා කළ යුතු දේ පිළිබඳ උපාය මාර්ගයක් මා වෙත යොමු කිරීමට උත්සහා දරමින් සිටියා.	1	2	3	4
15. මම කෙනෙකුගෙන් සැනසීම සහ තේරුම් ගැනීමක් ලබා ගනිමින් සිටියා.	1	2	3	4
16. සාර්ථකව මුහුණ දෙන්න උත්සහා කිරීම මම අහහැර සිටියා.	1	2	3	4
17. යම් දෙයක් සිදු වීමේදී හොඳ දෙයක් වෙයි කියා මම බලා සිටියා.	1	2	3	4
18. මම එය විනිඳුවක් ලෙස ගණන් ගනිමින් සිටියා	1	2	3	4
19. මම දුරටත් එය සිතිම අවම කර ගැනීමට විනුපටයක් නැරඹීමට යෑම,රූපවාහිනිය නැරඹීම,කියවීම,සිතින දැකීම,නිදා ගැනීම හෝ සාප්පු යෑම වැනි දේවල් මම කළා.	1	2	3	4
20. මම එය සිදු වී ඇති බව සහ සෑබෑ යථාර්ථය භාර ගන්නා.	1	2	3	4
21. මම මගේ සෘණාත්මක හැඟීම් ප්‍රකාශ කරමින් සිටියා.	1	2	3	4
22. මාගේ ආගම හෝ ආත්මික විශ්වාසයන් ගැන සැනසීම සොයා ගැනීමට මම උත්සහා කරමින් සිටියා.	1	2	3	4
23. මම කරන්න යන්නේ කුමක්ද ගැන සෙසු අයගෙන් උපදෙස් හෝ උපකාර ලබා ගැනීමට මම උත්සහා කරමින් සිටියා.	1	2	3	4
24. මම එය සමඟ පිවත් වීමට ඉගෙන ගනිමින් සිටියා.	1	2	3	4
25. ගත යුතු පියවර ගැන මම දැඩි ලෙස කල්පනා කරමින් සිටියා.	1	2	3	4
26. සිදු වුණු සමහර දෙවල් ගැන මම මටම චෝදනා කර ගන්නා.	1	2	3	4
27. මම යාවකදා කිරීම හෝ භාවනා කිරීමෙයි නිරත වුණා.	1	2	3	4
28. එම තත්වය ගැන විනෝදකාමි ලෙස සිතුවා.	1	2	3	4

3 වන කොටස :සංකෘතික විශේෂිත තත්වයන්

ණය ශ්‍රේණිය පහත පරිදි වේ:

1 = කිසිසේත්ම නැත

2 = ටිකක්

3 = මධ්‍යම ප්‍රමාණයේ

4 = බොහෝ සෙයින්

1. ශ්‍රී ලංකාවේ සිවිල් යුද්ධය ඔබට කොතරම් බලපෑවද?	1	2	3	4
2. 2004 වර්ෂයේ ඇතිවුණ සුනාමි තත්වය ඔබට කොතෙක් බලපෑවද?	1	2	3	4